



MIHIR SANJAYKUMAR PATEL

B.Tech. - Electronics & Communication Engineering

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering (ECE) student with hands-on experience in RTL design and constrained-random, coverage-driven functional verification using Verilog/SystemVerilog. Strong interest in computer architecture, cache coherency, and CPU microarchitecture, with a detail-oriented, spec-driven approach across the hardware design lifecycle. Collaborative team player eager to grow within a hardware engineering team.

KEY EXPERTISE

Skills: Verilog, SystemVerilog, ASIC, FPGA, RTL Design, Verification, DFT, ATPG, C, Python

Verification Methodologies: Layered SV Testbenches, Constrained-Random Stimulus, Coverage-Driven Verification

Soft Skills: Communication, Teamwork, Problem Solving

Tools: Cadence Genus/Modus/Xcelium, Xilinx Vivado, ModelSim, Quartus II, keil uVision, Microwind

EDUCATION

Institute of technology, Nirma University

2023 - 2027

B.Tech. - Electronics & Communication Engineering | CGPA: 8.70 / 10

New Meghdoot Vidya Vihar High School, Ahmedabad

2023

12th | GSEB | Percentage: 81.54 / 100

New Meghdoot Vidya Vihar High School, Ahmedabad

2021

10th | GSEB | Percentage: 94.50 / 100

Select Minor Subject : Computer Science

AWARDS AND SCHOLARSHIPS

- Received a 100% Tuition Fee Waiver (TFWS) Scholarship through merit-based selection for the top 5% based on Class 12 performance.

INTERNSHIPS

Synseis | Semiconductor

07 Jun, 2026 - Present

Edge AI Fellow

Key Skills: Verilog AI Digital Design SystemVerilog RTL Design Python

- Developed a 64-32-16-10 MLP hardware architecture using Posit-16 (es=1) for digit recognition and evaluated it against an equivalent FP32 implementation for hardware-oriented inference.
- Analyzed FPGA PPA reports, reducing LUT utilization from 6,231 to 2,800 (55.1% reduction) while maintaining comparable estimated on-chip power of 0.331 W versus 0.334 W for FP32.
- Implemented and verified Posit arithmetic datapaths and MLP inference using Verilog/SystemVerilog and Python, including hardware-oriented weight representation.

elinfochip - An Arrow Company | Semiconductor

15 May, 2026 - 30 Jun, 2026

ASIC intern

Key Skills: Embedded C Verilog Corporate Communications Presentation ESP32 SystemVerilog

- Built an ESP32-based smart-home system with six sensors and an SSD1306 OLED; created a self-hosted Wi-Fi HTTP dashboard in Arduino IDE.
- Executed ASIC physical design of a D flip-flop with OpenLane and Sky130 PDK, achieving timing and power closure and producing a verified GDSII file in KLayout.

PROJECTS

Design and the Verification of Asynchronous FIFO

Mentor: Dr. Vaishali Dhare | **Team Size:** 2

Key Skills:

Languages: Verilog, SystemVerilog EDA Tools: Vivado , EDA Playground

Methodologies: RTL Design, Functional Verification, CDC, Gray-Code Sync, Constrained-Random & Coverage-Driven Verification

Project Link: <https://github.com/MihirPatel-12/asy-fifo-verification>

Built a full verification environment for an asynchronous FIFO from the ground up, achieving 390/390 passing checks and 87.5–100% functional coverage across all covergroups, by implementing a layered SystemVerilog testbench with directed corner-case coverage and constrained-random stress testing, using separate generator, driver, monitor, and scoreboard layers with mailbox-based communication and virtual interfaces.

Scan Based DFT and ATPG Implementation of synchronous FIFO-using Cadence tools

Mentor: Dr. Vaishali Dhare | **Team Size:** 2

Key Skills:

Languages : Verilog, TCL EDA Tools : Cadence Genus, Cadence Modus, Xcelium Methodologies : ASIC, DFT, ATPG, Fault Coverage

Project Link: <https://github.com/MihirPatel-12/Scan-Based-DFT-and-ATPG-Implementation-of-synchronous-FIFO-using-Cadence-tools>

Achieved 99.98% adjusted stuck-at fault coverage (99.97% test coverage) on a synchronous FIFO design, by implementing a full scan-based DFT and ATPG flow with muxed scan insertion, automatic scan chain stitching, and stuck-at ATPG pattern generation using Cadence Genus and Modus.

RISC-V Processor with Custom AES Instruction Set

Mentor: Dr. Dhaval Shah | **Team Size:** 2

Key Skills: Languages : Verilog, SystemVerilog Tools : Xilinx Vivado Methodologies : RTL Design, RISC-V, Cryptographic Hardware

Project Link: <https://github.com/MihirPatel-12/RISC-V-Processor-with-Custom-AES-Instruction-Set>

Achieved single-cycle (1 clock cycle) latency for full AES-128 encryption, validated correct via the FIPS-197 known-answer test, by designing a custom RISC-V instruction-set extension with a fully unrolled 10-round hardware AES-128 datapath in Verilog HDL, verified through cycle-accurate simulation in Xilinx Vivado.

ESI Cache Coherence Controller (Verilog)

Mentor: Dr. Dhaval Shah | **Team Size:** 2

Key Skills:

Languages: Verilog EDA Tools: EDA Playground

Methodologies: RTL Design, FSM Design, ESI Cache Coherence, Functional Verification, Reference-Model Checking, Fault Injection

Project Link: https://github.com/MihirPatel-12/Esi_Cache_Controller

Designed and verified a synthesizable ESI cache coherence controller in Verilog, achieving 100% pass rate across 66 assertions spanning 31 test scenarios, by implementing a single-cycle-latency FSM with snoop-priority arbitration and validating it against an independent software reference model.

ASSESSMENTS / CERTIFICATIONS

Foundations of VLSI Design Verification

Provider: Electronics & ICT Academies MNIT Jaipur, IIITDM Jabalpur, IIT Roorkee & NIT Patna

Key Skills: SystemVerilog Verilog RTL Design Digital Design

System Design Through Verilog

Provider: IIT Guwahati

Aggregate: 81 / 100

Key Skills: Verilog RTL Design Digital System Design

WEB LINKS

- Github - <https://github.com/MihirPatel-12>